Acoustic features only (all 40 features)

FINAL_V1 experiment group summary

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Results

- 4 of 4 conditions in this group are complete.
- Highest pooled AUPRC (descriptive, selection-biased among 4 conditions): AC Logistic regression, AUPRC 1.000, AUROC 1.000.
- Lowest: AC RBF SVM, AUPRC 0.991, AUROC 0.989.
- Range of AUROC over the group: 0.989 to 1.000; median 0.999.
- Speaker errors at threshold 0.5 (false negatives + false positives) per condition: minimum 1, median 1, maximum 4 (out of 47 speakers).
- Fold variability: median over conditions of the mean fold AUROC is 1.000 with a median SD of 0.000 over the five outer folds (9 to 10 speakers per fold).

Reading the table

- Primary metrics are pooled speaker-level out-of-fold AUPRC and AUROC (47 speakers: 25 PD, 22 controls; prevalence 53.2% PD); F1, sensitivity, specificity and balanced accuracy use the fixed threshold 0.5.
- Intervals are percentile bootstrap intervals over speakers (2,000 resamples); with 47 speakers one person moves a metric by roughly 2 points. Comparisons are descriptive; no significance is claimed.
- Recording date and class are confounded in this corpus (every date holds one class; all 44.1 kHz audio is PD), so a high score does not show disease detection.

Table 1: Pooled out-of-fold speaker-level results

Representation	Model	AUPRC [95% CI]	AUROC [95% CI]	F1	Sens.	Spec.	Bal.acc.
AC	Logistic regression	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
AC	Linear SVM	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
AC	RBF SVM	0.991 [0.971, 1.000]	0.989 [0.964, 1.000]	0.923	0.960	0.864	0.912
AC	XGBoost	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980